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Survey of Ad Hoc Capital Injections[1]

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Yale Program on Financial Stability Survey
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Abstract

Government recapitalizations of systemic banking organizations can be costly and unpopular but are sometimes necessary to protect depositors and prevent financial contagion. This paper surveys 23 Yale Program on Financial Stability case studies of ad hoc capital injection programs, defined as programs that provide capital to a single institution or a clearly defined minority of institutions. We saw a marked increase in capital injection programs in the past 50 years, more than half of which were ad hoc. Authorities designing ad hoc capital injections face difficult decisions—when to deploy them rather than let a bank fail; whether to impose losses on shareholders and unsecured creditors; how to protect taxpayers; and how to deal with the management of failed firms. Most advanced countries have updated their tools for managing bank failures since the Global Financial Crisis of 2007–09, generally in compliance with the Financial Stability Board's 2011 Key Attributes of Effective Resolution Regimes for Financial Institutions. Our survey points to the challenges authorities have faced over the years in attempting to adhere to such principles.

Keywords: ad hoc, bail-in, bank capital, capital injection, recapitalization

[1] This survey reviews a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering ad hoc capital injection programs. It references an earlier survey of broad-based capital injection programs (Rhee et al. 2022). These surveys and the individual cases underlying them are available from the *Journal of Financial Crises* at https://elischolar.library.yale.edu/journal-of-financial-crises/.

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Introductory Note: This survey is an analysis of important considerations for policymakers seeking to establish an ad hoc capital injection (AHCI) program. It is based on insights derived from case studies of 23 specific AHCI programs the Yale Program on Financial Stability has completed and from the existing literature on the topic. While this survey can help inform a decision about whether or not to establish an AHCI program, our main purpose is to assist policymakers who have already made that decision in designing the most effective program possible. In analyzing the programs that are the focus of this survey, we used a color-coded system to highlight certain particularly noteworthy design features.

Treatment	Meaning
BLUE – INTERESTING	A design feature that is interesting and that policymakers may want to consider. Typically, this determination is based on the observation that the design feature involves a unique and potentially promising way of addressing a challenge common to this type of program that may not be obvious. Less commonly, empirical evidence or a consensus will indicate that the design feature was effective in this context, in which case we will describe that evidence or consensus.
YELLOW – CAUTION INDICATED	A design feature that policymakers should exercise caution in considering. Typically, this determination is based on the observation that the designers of the feature later made significant changes to the feature with the intention of improving the program. Less commonly, empirical evidence or a consensus will indicate that the design feature was ineffective in this context, in which case we will describe that evidence or consensus.
FOOTNOTE IN ITALICS	Where the reason that a given design feature has been highlighted is not apparent from the text, it is accompanied by an italicized footnote that explains why we chose to highlight it. Where necessary, these footnotes will be used to identify any considerations that should be kept in mind when thinking about the feature.

This highlighting is not intended to be dispositive. The fact that a design feature is not highlighted or is highlighted yellow does not mean that it should not be considered or that it will never be effective under any circumstances. Similarly, the fact that a design feature is not highlighted or is highlighted blue does not mean that it should always be considered or will be effective under all circumstances. The highlighting is our subjective attempt to guide readers toward certain design features that (1) may not be obvious but are worth considering or (2) require caution.

Introduction

Government recapitalizations of systemic banking organizations can be costly and unpopular, but authorities sometimes decide they are necessary to protect depositors and other creditors and prevent financial contagion. This survey covers 23 ad hoc capital injection (AHCI) cases, defined as programs that provided capital to a single institution or a clearly defined minority of institutions: two before World War II; two in the Asian Financial Crisis of 1997–98; 11 during the Global Financial Crisis (GFC) of 2007–09; six in the post-GFC European Union (EU); and two post-GFC cases outside the EU. In 2022, we published a survey on broad-based capital injection (BBCI) programs, defined as programs that addressed all or most of a banking system to address a systemic banking crisis (Rhee et al. 2022). The Metrick and Schmelzing (2024) database analyzes 1,946 banking crisis interventions from 1257–2019 across seven intervention categories. The database shows a marked increase in capital injections since 1972: they accounted for only 67 (6.8%) of the 984 interventions before 1972 compared to 221 (23%) of the 962 interventions since 1972—representing the most commonly used intervention tool in this period (1972–2019). More than half of the capital injection programs identified were ad hoc. Additionally, the BBCI cases we studied largely had voluntary participation. In contrast, participation in all of the AHCI cases was mandatory; typically, the target bank required a capital injection to survive. This difference influences different designs in the programs, which we will discuss further in the survey.

Figure 1: Case Studies and Abbreviated Names

Case Name	Short-form Case Name	Reference
Austria and Germany: Hypo Alpe Adria Capital Injection, 2008	Austria and Germany, HAA	(Decker 2024a)
Belgium and France: Dexia Group Capital Injection, 2008	Belgium and France, Dexia	(George 2024a)
Belgium: Fortis Group Capital Injection, 2008	Belgium, Fortis	(George 2024b)
Cyprus: Laiki Bank Capital Injection, 2012	Cyprus, Laiki	(Schaefer-Brown 2024a)
France: Bank of France Capital Injection, 1805	France, Bank of France	(George 2024c)
Germany: HSH Nordbank Capital Injection, 2009	Germany, HSH	(Makhija 2024a)
Germany: IKB Deutsche Industriebank Capital Injection, 2008	Germany, IKB	(Swaminathan and Vala 2024)
Greece: ATE Bank Capital Injection, 2011	Greece, ATE	(Schaefer-Brown 2024b)
Greece: Piraeus Bank Capital Injection, 2015	Greece, Piraeus	(Schaefer-Brown 2024c)
Iceland: Arion Bank, Islandsbanki, and Landsbankinn Capital Injection, 2008	Iceland	(George 2024d)
India: Yes Bank Capital Injection, 2020	India, Yes Bank	(Gupta 2024)
Italy: Banca Monte dei Paschi di Siena Capital Injection, 2017	Italy, MPS	(Hoffner 2024a)
Japan: Nippon Credit Bank Capital Injection, 1997	Japan, NCB	(Heaphy 2024)
Korea: Korea First Bank and Seoul Bank Capital Injection, 1997	Korea, KFB and SB	(Park and Heaphy 2024)
Latvia: Parex Bank Capital Injection, 2008	Latvia, Parex	(Decker 2024b)
Netherlands: SNS Reaal Capital Injection, 2013	Netherlands, SNS Reaal	(George 2024e)
Portugal: Banco Espírito Santo Capital Injection, 2014	Portugal, BES (Novo Banco)	(Gupta and Brandon 2024)
Russia: Otkritie Bank Capital Injection, 2017	Russia, Otkritie	(Hoffner 2024b)
Spain: Caja de Ahorros Castilla–La Mancha Capital Injection, 2009	Spain, CCM	(Swaminathan 2024)
Switzerland: Schweizerische Volksbank Capital Injection, 1933	Switzerland, Volksbank	(Makhija 2024b)
Switzerland: UBS Capital Injection, 2008	Switzerland, UBS	(Makhija 2024c)
United States: Bank of America Capital Injection, 2009	US, BofA	(Hoffner and Arnold 2024a)
United States: Citigroup Capital Injection, 2008	US, Citi	(Hoffner and Arnold 2024b)

Source: Authors' compilation.

In most AHCI cases, governments applied some basic principles to recapitalizations, including raising private capital first, imposing losses on private investors, promoting new management, improving oversight, etc. Since the GFC, these basic principles have been incorporated into regulations across the world. In 2011, regulators in advanced economies agreed to Key Attributes of Effective Resolution Regimes for Financial Institutions to make bailouts less likely and less costly (Financial Stability Board 2014). The key attribute on funding of firms in resolution states principles around authorities providing temporary funding to banks, including a goal of preserving financial stability and allocation of losses to stakeholders.

These key attributes were incorporated most notably by the European Union, with the adoption of the Bank Recovery and Resolution Directive (BRRD) in 2014. The BRRD established a common framework for the resolution of failing banks and lays out general principles for capital injections outside and inside resolution. For capital injections outside resolution, the BRRD requires that aid should be confined to solvent institutions and be of a precautionary and temporary nature. Another principle, generally applying to both situations, includes the use of a bail-in tool, ensuring that the bank's losses are first borne by shareholders and creditors.

The International Monetary Fund (IMF) published a paper in 2020 summarizing general principles for managing systemic banking crises, based on the views and experience of its staff. Its principles around government capital injections were: (1) Recapitalizations should be made only to address systemic risks; (2) private shareholders and unsecured creditors should share in the loss burden; (3) public taxpayers should be remunerated and share in any upside from a bank's recovery; (4) management should be replaced, and some jurisdictions also could seek to have compensation capped and bonuses clawed back; (5) authorities should maintain strict oversight of recapitalized firms to protect taxpayers and establish independent oversight and ex post reviews; and (6) the authorities should be transparent about losses to taxpayers and recoveries (Moretti, Chavarri, and Dobler 2020).

These principles and regulations provide a road map for some of the most important design decisions authorities face in planning capital injections. However, a review of our 23 AHCI and 36 BBCI case studies suggests nuances in that road map. In the BBCI survey, we emphasized that distinguishing between the goals of BBCI programs implemented during acute ("panic") and chronic ("debt overhang") phases of a crisis is crucial. In an acute phase, we found that BBCI programs should be designed to influence the behavior of bank counterparties, while in chronic phases, the focus should be on bank behavior itself. We observed some parallels with this distinction in the ad hoc cases, but with acute and chronic defined at the entity level. When authorities are motivated by liquidity concerns—in the midst of a run or in fear of one—we consider that an acute-phase intervention, even if the overall financial system is not in danger. On the other hand, when authorities are worried more about long-run viability than short-term liquidity, we consider that to be a chronic-phase intervention.

We identified several themes to guide capital injection program design in the BBCI survey and provided many illustrative examples for the policymaker's tool kit. The lessons from our BBCI survey that overlap with this AHCI survey include ensuring that: (1) the size of a capital injection be "large enough" to solve the problem without need for further injections; (2) capital injections be combined with other interventions; (3) entities beware of legal or regulatory constraints on program implementation; (4) the exit plan fits the investment type; and (5) in the acute phase, higher-tier capital is better. These themes were strengthened by our analysis of the 23 AHCI cases.

Additionally, our survey on restructuring and resolution drew lessons from 19 case studies of bank resolutions, 10 of which included recapitalizations (McNamara et al. 2024). These lessons were: (1) the need for resolution and restructuring to eliminate uncertainty about an institution's solvency by closing it, recapitalizing it, or merging it with a healthier institution; (2) the importance of effective valuation in achieving this result; (3) the necessity of clarity in the treatment of creditors; and (4) the value of a credible bail-in tool to incentivize creditors to agree to solutions outside resolution. This survey provides further evidence of these themes.

Although capital injections can be costly, both financially and politically, authorities often find that they are necessary to protect depositors and prevent financial contagion. Authorities should be aware of regulatory constraints on program implementation. This was especially true for AHCI cases in terms of writing down or bailing in existing stakeholders as a condition for capital injection. As emphasized in the BBCI survey, policymakers are encouraged to address as many of these problems as possible through legislation passed in quieter times before a crisis occurs, allowing for emergency authority to be utilized efficiently. With this overarching recommendation, we list and discuss four themes for institutions contemplating an ad hoc capital injection program.

The major challenge in making capital injections is balancing the tension between financial stability measures and the political backlash and moral hazard they engender. Moral hazard concerns often drive punitive ex ante conditions for capital injections; yet, financial stability concerns, focused on the spillover effects of a large bank failure on individuals and markets, should outweigh moral hazard concerns in the acute phase of a financial crisis. Authorities should instead address the political backlash and moral hazard concerns with tough measures such as bail-in—imposing losses on shareholders and creditors—and discipline for management, including through legal channels if laws are broken. But authorities often balk at such measures in the depths of a crisis. Especially for bail-ins, even with appropriate legal authority, clarity and consistency are important factors in implementing the bail-in or write-down of existing stakeholders. Unexpected losses imposed on stakeholders can lead to panic in the wider market and multiple lawsuits against the state by stakeholders seeking compensation.

In contrast, in a chronic situation, there may be less concern that such a surprise could spook investors, and authorities may be more able to impose punitive outcomes on stakeholders.

Even in cases where substantial taxpayer losses are unavoidable, authorities can use various tools to enable taxpayers to share some upside from a capital injection. This can be done by ensuring the state holds equity or warrants in a bank as it attempts to recover. Taxpayers can also be protected through strong oversight of the bank after the capital injection. In the US AHCI cases for Citi and BofA, the law authorizing capital injection required authorities to acquire warrants for the right to receive nonvoting common shares or preferred shares of the financial institution for the benefit of taxpayers.

Combining capital injections with other interventions is good practice. Combining other programs with a capital injection can be a useful force multiplier, as we noted in the

BBCI survey. Fourteen of the 23 cases we studied also included guarantee programs (as noted in Key Design Decision No. 2, Part of a Package). The type of guarantee varies depending on the failing bank's situation. Guaranteeing a bank's liabilities may help stem runs by anxious creditors and depositors, but it may not convince anyone that the bank itself is solvent. For that reason, in many cases, governments have also guaranteed a bank's assets, capping the potential for further credit or other losses.

Exit plans are important, and capital design can help with the exit. From the start of the capital injections, authorities generally intend to reprivatize their investments as soon as possible. Principles and regulations around recapitalization also emphasize the temporary nature of the government support through capital injection. Also, as we noted in the BBCI survey, the absence of a feasible exit plan can lead to higher costs, both politically and economically.

Key Design Decision No. 8, Capital Characteristics, and Key Design Decision No. 15, Exit Strategy, discuss examples helpful in accomplishing the exit.

Governments may face constraints that render conventional cash-based recapitalizations difficult. Using government bonds in such instances can work, provided that the sovereign has stable credit status. These bond-based recapitalizations were a common practice in financial crises before 2000 (Andrews 2003), although we noted in our own survey on BBCI programs that they have been less common since the GFC (Rhee et al. 2022). Governments may choose bond-based recapitalizations because their domestic bond markets are relatively undeveloped, but even countries with developed markets have chosen this path because of the technical or political constraints associated with financing recapitalizations. Of course, injecting cash is typically preferred to issuing government debt in return for new bank capital in a jurisdiction where the sovereign debt rating has become unstable. In those instances, quickly deteriorating or questionable value of government bonds may further strain the bank's viability. Key Design Decision No. 9, Size and Source of Funding, discusses the issue in further detail.

The remaining sections of this paper go through the specific Key Design Decisions in detail. As we did in the BBCI survey, we note at the outset of this analysis that most of these design decisions were made in real time, often during the most acute phases of a crisis. In almost all cases, the legal background for the programs was either new or untested, and many mistakes were made because of this time pressure and novelty. In recent years, most countries with significant banking systems have improved their legal frameworks around bank resolutions to limit the need for official recapitalizations and to tightly circumscribe the conditions of their use. The global regulatory community has also reached a consensus about the need for large banks and other financial institutions to maintain plans for their own resolution, overseen by their regulators or resolution authorities. The survey allows for a more nuanced approach to those principles and regulations depending on the situation a policymaker is facing. A metatheme of much of the work of the Yale Program on Financial Stability is that program design could be significantly improved if policymakers worked out some of these details in advance.

Key Design Decisions

1. Purpose: What was the policymaker's justification and goal for the capital injection to the institution?

All cases in our survey can largely be divided into having two objectives. One is a capital injection to a distressed financial institution to prevent it from becoming nonviable, and the other is a capital injection to rebuild the institution for longer-term viability as a private concern. The former objective can also vary from a truly preemptive intervention to prevent imminent bankruptcy of the institution, and the goal is to address bank counterparties. Both objectives accompany an emphasis from the policymaker on the systemic importance of the institution to the banking sector, financial system, and the entire economy.

Ultimately, recapitalization should be for viable institutions regardless of the purpose of the intervention. For nonviable institutions, authorities should choose resolution, restructuring, or if there is a compelling national interest, nationalization. However, determining viability is a challenge during a financial crisis, and authorities can't always make the right call. The resolution of Dunfermline Building Society is an example of a case in which the authorities ultimately decided not to inject capital, after balancing the options of recapitalization and resolution. When the official stress tests did not persuade policymakers that the institution would be able to service the interest on its injected capital or repay it, they elected to put the institution through special resolution under its banking law (Makhija 2024d). In contrast, in the Laiki case in Cyprus, although the government's decision to recapitalize the bank was based on a viability assessment, a later study shows that this assessment was in error, and the European Central Bank (ECB) further criticized the Cypriot authorities for recapitalizing a nonviable bank. The ECB recommended instead using resolution tools (Schaefer-Brown 2024a). In the MPS case in Italy, an academic has argued that the bank should not have been eligible for public recapitalization based on its failing to meet long-term viability criteria under BRRD and Single Resolution Mechanism (SRM) rules (Hoffner 2024a).

Sometimes, policymakers choose capital injection because of the downside of other intervention options. In the case of Austria's HAA, when it turned out that the bank needed further capital after the first injection, instead of resolution, the policymakers decided to go ahead with a second injection leading to nationalization as there were no suitable laws to deal with the bank's bankruptcy at that time. In addition, the policymakers had several concerns about bankruptcy since the bank had a guarantee from a small Austrian state and bankruptcy of the bank would have in turn bankrupted this state. Further, the bank had a substantial presence in Central, Eastern, and Southeastern European countries, all of which were facing deteriorating economic situations at that time and receiving or in the process of negotiating support from the IMF, the European Commission (EC), and the World Bank. The bank's bankruptcy not only would have further negatively impacted these countries but also would have led the regulators in these countries to ring-fence certain assets of the bank, causing its overall asset value to deteriorate. Other Austrian banks with a major presence in these countries may also have been negatively impacted by association (Decker 2024a).

In the case of Germany's HSH, the authorities had similar concerns to HAA, where a resolution would have bankrupted a regional government because of the guarantee it was providing to the bank. Thus, the authorities decided to inject capital.

2. Part of a Package: Did the policymakers announce the capital injection alongside other interventions? Were there any intervention programs existing when the capital injection took place?

All the instances of ad hoc capital injection we studied accompanied other intervention tools. In many instances, policymakers had already provided emergency liquidity. Switzerland's Volksbank, Spain's CCM, and Korea's KFB and SB all had received liquidity, but losses continued to increase, and authorities had to inject capital. In Russia's Otkritie, the central bank provided liquidity enabling the bank to stay operational before receiving capital injections. Fourteen of the cases had a guarantee accompanying the capital injection. There were three instances of using moratorium to stop runs in addition to capital injection. Nine of our cases were open-bank transactions, in which the government never had to resolve the institution. All of the other AHCI cases accompanied or resulted in the resolution or restructuring of the institution. In 11 of the 23 cases surveyed, the authorities ultimately put the financial institution receiving the aid into resolution. In 12 cases, the authorities found private sector buyers. In Otkritie, a state bank bought the authorities' stake in the bank. In all of these cases, the government declared its intent to reprivatize as soon as possible. Figure 2 lists different combinations of policy measures in operation with the AHCI.

Figure 2: Other Policy Measures with Capital Injection

Case Name	Emergency Liquidity Assistance	Guarantee	Resolution and Restructuring	Other Programs
Austria and Germany, HAA	✓	Account, asset, debt	✓	
Belgium and France, Dexia	✓	Debt	✓	
Belgium, Fortis	✓		✓	
Cyprus, Laiki	✓		✓	
France, Bank of France				Moratorium
Germany, HSH		Asset, debt		
Germany, IKB	✓	Asset, debt		
Greece, ATE	✓	Debt	✓	
Greece, Piraeus				
Iceland	✓	Account	✓	
India, Yes Bank	✓		✓	Moratorium
Italy, MPS	✓	Debt		
Japan, NCB	✓		✓	
Korea, KFB and SB	✓	Debt	✓	
Latvia, Parex	✓	Debt	✓	Moratorium
Netherlands, SNS Reaal	✓	Debt	✓	
Portugal, BES (Novo Banco)			✓	
Russia, Otkritie	✓		✓	
Spain, CCM	✓	Asset	✓	
Switzerland, Volksbank	✓			
Switzerland, UBS	✓	Account		
US, BofA	✓	Account, asset, debt		
US, Citi	✓	Account, asset, debt		

Source: Authors' compilation.

As studied in the BBCI survey, bank restructuring and asset and debt guarantees are complementary to recapitalization in AHCI. While capital injections are the most direct method to solve an undercapitalization problem, they are both fiscally and politically expensive. A guarantee program, on the other hand, incurs only contingent liabilities for the government, and the absence of immediate outlays can make it politically more palatable.

Of course, the motivation for guaranteeing a bank's liabilities (accounts or other debt) may differ from the motivation for guaranteeing its assets. A liability guarantee may address a liquidity problem but not a solvency problem. In other words, it may convince liability holders not to run, but it may not persuade anyone that the bank itself is solvent. Ten cases included liability guarantees; four of those also had asset guarantees; and two cases had only asset guarantees. In the HSH case, the authorities, after guaranteeing banking debt, created a model for interventions to ensure the bank could meet its regulatory capital requirement while minimizing the budgetary impact by combining a capital injection and an asset guarantee to reduce the bank's risk-weighted assets (Makhija 2024a).

Similarly, during the Global Financial Crisis, the US bank deposit insurer, the Federal Deposit Insurance Corporation (FDIC), guaranteed senior debt and transaction accounts for all banks to stop widespread runs, and the Treasury injected capital into all of the largest banks, including Citi. However, market concerns continued to focus on the low quality of Citi's assets and the threat to its solvency. In response, the authorities constructed a package that consisted of both a second round of public sector capital and an asset guarantee. Under the asset guarantee, the government and central bank would share in the losses on a large pool of distressed assets if those losses proved to be catastrophic. The authorities later negotiated a similar deal for BofA, as market concerns similarly focused on the quality of assets at Merrill Lynch, the investment bank that BofA had just acquired with the government's blessing. As it turned out, this effort still wasn't sufficient to restore confidence in major US banks. The authorities later ran the largest banks, including Citi and BofA, through a stress test and swapped some of their capital for higher-quality common equity to further reassure market participants of the banks' solvency.

3. Legal Authority: What was the legal basis that the authorities relied on? Was there a need for new law, or could they rely on existing authority? For European Union member countries, what extra European Commission approval did they require?

As we found in the BBCI survey, many jurisdictions required a new regulatory framework to inject capital into the institutions in the ad hoc cases studied. In 15 cases, the capital injections relied on new legal regimes. This meant that some authorities needed to be creative in stretching existing regimes to find a basis for a timely capital injection.

In the UBS case, Switzerland relied on an interpretation of existing government authority to issue ordinances to protect the country's foreign relations and security, and the Swiss Parliament approved the capital injection retroactively (Makhija 2024c). The government reasoned that without intervention, UBS's failure would bring civil disturbance and reputational damage to the Swiss financial market. However, subsequent legal analysis has noted that this authority was intended only for such extraordinary and unforeseeable situations as natural disasters, epidemics, and military threats, and although the collapse of a major bank would have led to some civil disturbance, the prevention of the collapse was likely an insufficient basis to issue the ordinance. Therefore, without the retroactive parliamentary approval, the repercussions the government would have faced are unclear.

4. Administration: Which authority was in charge of establishing and operating the capital injection?

Typically, the fiscal authority has taken the lead in establishing and administering the recapitalization of financial institutions. This is consistent with our observation in the BBCI survey and the recommendation by the IMF in managing a systemic banking crisis. The IMF notes that ownership interests in banks after recapitalization should be with the finance ministry or a specialized agency. It further states that central banks, other supervisors, and deposit insurers are inadequate. This is because they have conflicts of interest as owners of the banks, and the ownership can undermine the credibility and balance sheets of central banks or deposit insurers (Moretti, Chavarri, and Dobler 2020). Often, in our AHCI cases, when the fiscal authority took the lead, central banks and other supervisory and regulatory agencies assisted in the operations. Figure 3 lists the main operators in our 23 AHCI cases.

Figure 3: Administrator Responsible for Capital Injection Operations

Case Name	Administrator
Austria and Germany, HAA	Special entity (FIMBAG)[a]
Belgium and France, Dexia	Special entity (SFPI)[b]
Belgium, Fortis	Special entity (SFPI)[b]
Cyprus, Laiki	Government
France, Bank of France	Government
Germany, HSH	Government
Germany, IKB	State-owned development bank
Greece, ATE	Government
Greece, Piraeus	Special entity (HFSF)[c]
Iceland	Government and supervisor
India, Yes Bank	Central bank
Italy, MPS	Government
Japan, NCB	Government
Korea, KFB and SB	Central bank and deposit insurer
Latvia, Parex	Government and special entity (LPA)[d]
Netherlands, SNS Reaal	Government
Portugal, BES (Novo Banco)	Special entity (resolution fund)
Russia, Otkritie	Central bank
Spain, CCM	Deposit insurer
Switzerland, Volksbank	Government
Switzerland, UBS	Government
US, BofA	Government
US, Citi	Government

(a) FIMBAG: subsidiary of Austria's sovereign wealth fund.

(b) SFPI: Federal Participation and Investment Company, a quasi-private entity, under the direction of the Belgian government (with support of French and Luxembourgish authorities, as applicable).

(c) HFSF: Hellenic Financial Stability Fund.

(d) LPA: Latvian Privatization Agency.

Source: Authors' compilation.

The central bank or the deposit insurer has taken the lead in a smaller number of cases. In the case of Yes Bank in India, the central bank coordinated and administered the capital injection by a consortium of banks (Gupta 2024). For Russia's Otkritie, the central bank formed a special vehicle to administer the capital injection, and all ownership rights were exercised with approval from the central bank (Hoffner 2024b). In the case of Japan's NCB, the central bank administered the capital injection, and the Ministry of Finance was unofficially involved in coordinating any private consortium efforts (Heaphy 2024). In the KFB and SB case, the central bank originally administered the capital injection, but then later the deposit insurer took over the role (Park and Heaphy 2024). The deposit insurer fully administered the capital injection in the case of Spain's CCM (Swaminathan 2024).

5. Governance: Were any bodies overseeing the administration of capital injection? Were there requirements to report to a legislative body?

We discussed in the BBCI survey that in devising governance arrangements, authorities must pay attention to both the operational efficiency of the program and the public perception of its fairness. The most common form of governance was the regular reporting obligation to the parliament. In the case of KFB and SB, Korea had created the Public Fund Oversight Committee to oversee the recovery of the funds, and authorities were mandated to regularly disclose on how the funds injected to KFB and SB were being managed (Park and Heaphy 2024). EU countries' programs have an obligation to report to the EC on the operations of the program, and for certain cases, the EC assigns a monitoring trustee to oversee the operations.

6. Communication and Disclosure: How did the government communicate the goals of the program? Were there any special communication issues?

Policymakers strive to quickly reinstall confidence in the banks and the financial system through the announcement of capital injections. Successful communications are clear about the goals of the program, send consistent messages, and portray the authorities as competent and decisive (Rhee et al. 2022). Any sudden change in the plans of the authorities will cause confusion. In Otkritie's case, Russia's central bank initially announced that the creditors of the bank would not be bailed in, but the next day, the central bank flipped the decision, saying it may implement a bail-in depending on the bank's capital level.

A leak that a bank may be receiving a capital injection before an official communication by the policymakers can worsen the financial situation of the bank. When the news of Volksbank's difficulties and bailout had leaked a day before Swiss policymakers announced the capital injection, Volksbank's liquidity drained in the following days as worried depositors withdrew their savings. The federal loan office and the central bank had to provide emergency loans to bridge payment difficulties until the capital injection was completed (Makhija 2024b).

In Korea, the letter of intent sent to the IMF by the central bank and the Ministry of Finance stated that "[t]wo commercial banks in distress are required to prepare a plan, in about two months, to meet the Basel capital standards within four months after approval of the plan"

(IMF 1997). When the letter was disclosed, which is standard practice for the IMF, speculation that the two banks unnamed in the letter were KFB and SB grew and caused a massive run on both banks (Park and Heaphy 2024).

7. Treatment of Creditors and Equity Holders: How were the claims of creditors and equity holders treated during capital injection?

Thirteen of our AHCI cases included the practice of imposing losses on certain stakeholders of the bank. Of those, 10 required banks' equity shareholders to absorb losses before the government capital injection; eight imposed losses on subordinated debt or preferred shareholders; and one (Laiki) imposed losses on uninsured depositors. In contrast, participation in most of the BBCI programs we examined in our earlier survey was voluntary and, for that reason, did not impose losses on stakeholders; a minority of those programs were mandatory and did impose losses at least on equity shareholders. As in the context of the resolution, how to deal with an institution's loss is a major question for AHCI. This section will have many interconnected discussions with McNamara et al. (2024), a survey on resolution and restructuring. Figure 4 shows on whom the losses were imposed for each case.

Figure 4: Imposition of Losses on Stakeholders

Case Name	Bail-in
Austria and Germany, HAA	Equity holders
Belgium and France, Dexia	None
Belgium, Fortis	None
Cyprus, Laiki	Equity holders, bondholders, uninsured depositors
France, Bank of France	None
Germany, HSH	None
Germany, IKB	None
Greece, ATE	None
Greece, Piraeus	Noteholders
Iceland	Equity holders, bondholders
India, Yes Bank	Equity holders, AT1 bondholders
Italy, MPS	Subordinated bondholders
Japan, NCB	Equity holders
Korea, KFB and SB	Equity holders
Latvia, Parex	Equity holders
Netherlands, SNS Reaal	Equity holders, subordinated bondholders
Portugal, BES (Novo Banco)	Equity holders, subordinated bondholders, senior bondholders
Russia, Otkritie	Subordinated bondholders
Spain, CCM	None
Switzerland, Volksbank	Cooperative shareholders
Switzerland, UBS	None
US, BofA	None
US, Citi	None

Source: Authors' compilation.

Although many discussions around bail-ins and loss-sharing seem recent, our cases show that this practice existed in 1933 with Volksbank and during the 1990s in Korea and Japan. The IMF, in a 2020 report on lessons learned from managing systemic banking crises, recommended that before recapitalizing a bank with public funds, the losses of the bank must first be recognized. This requirement prevents public funds from bailing out equity shareholders, who are intended to absorb losses when a firm becomes troubled (Moretti, Chavarri, and Dobler 2020). Imposing losses on these stakeholders is based on the principle that the costs of bank failures are shifted to where they best belong. This approach is also to replace the public subsidy with private penalty or with private insurance forcing banks to internalize the cost of risks they assume (Goodhart and Avgouleas 2015).

This practice of burden sharing by equity shareholders is further supported by the European Commission in the cases of Austria's HAA, Latvia's Parex, and Germany's HSH. The Commission emphasized sufficient bail-in reflecting the true losses of the bank in approving capital injections under the State Aid rules. Authorities in the HSH case had not imposed losses on the existing shareholders before the capital injection, and the Commission disagreed with this practice, arguing the owners should have been completely diluted before the injection. It took two years for the Commission to deliberate on the State Aid decision, and in its final decision, it required the bank to pay an additional EUR 500 million in shares to the authorities. Following this final decision, the bank's shareholders appealed for an annulment of the decision in the EU General Court. However, the General Court dismissed the appeal since the Commission's decision created proper burden sharing (Makhija 2024a).

Burden sharing can also affect stakeholders beyond equity shareholders. Eight cases imposed losses on preferred shareholders or subordinated debt holders. The Commission criticized Spanish authorities in the CCM case for allowing remuneration for certain hybrid capital holders. CCM called two previously issued preference shares at par value after the capital injection despite the absence of distributable profits. Also, CCM had made discretionary coupon payments on hybrid capital although it was incurring losses in the previous year (Swaminathan 2024). The EC noted that this was inconsistent with its policies on hybrid instruments and its principles on burden sharing since the creditors were not made to bear any of the losses. This was despite CCM having no legal obligation to do this and the central bank having the right to bail in the debt before any recapitalization of the entity (Santos 2017).

In the case of Cyprus's Laiki, policymakers called for voluntary bail-in through exchange of perpetual subordinated debt and subordinated bonds, respectively, into new ordinary shares and new subordinated bonds with a discount. This approach was criticized for not being aggressive enough (Schaefer-Brown 2024a). Dübel (2013) asserts that Laiki would have had more of a risk buffer if it had written down shareholders and hybrid capital owners to zero and turned subordinated bond investors into co-shareholders at the time of the government recapitalization. In addition, Dübel (2013) argues that by writing down uninsured depositors, policymakers could have protected themselves against losses. In 2013, the government resolved Laiki, and most uninsured deposits were bailed in as they remained in the legacy bank as it was liquidated over time. This, however, did undermine the confidence in the Cypriot banking sector (McNamara et al. 2024).

In some instances, products intended to absorb losses end up in the wrong hands. In the MPS case in Italy, the recapitalization included the mandatory conversion of subordinated bonds into new equity shares. However, these subordinated bonds had been sold to retail investors before the bank got into trouble, and the authorities ultimately set up a fund to compensate for these retail investors after the write-down of all subordinated bonds (Hoffner 2024a). Authorities may be also reluctant to impose losses on certain creditors because of the unique situations of the creditor. Initially, Austrian authorities did not allow any haircuts on HAA's senior bonds guaranteed by one of its states, as losses on this state would have hurt Austria's overall reputation as a borrower (Decker 2024a). Although this decision may have been appropriate in the moment, we note that it led to unintended consequences: the existing state guarantee subsequently tied the authorities' hands in effectively deploying crisis-fighting tools. When the authorities ultimately resolved the bank, the state guarantee complicated the resolution and incurred significant public costs (McNamara et al. 2024). In the Netherlands' SNS Reaal case, authorities intentionally preserved the senior creditors of the bank to prevent rating downgrades of the bank and because of concerns around risks of negative impacts on the cost and stability of funding for the rest of the Dutch banking sector (George 2024e). However, shareholders and subordinated creditors were wiped out, and a large number of them brought lawsuits against the government. Eight years after the capital injection and the bail-in, the court ordered the government to provide compensation to certain subordinated creditors, and two years after the court order, the government started a compensation scheme.

Ex ante, having the bail-in regime in place is important, especially for nonequity stakeholders. Without appropriate legal authorities bestowed upon the policymakers, it took more than three months for Russian authorities in the Otkritie case to write down subordinated creditors (Hoffner 2024b). Though initially stating that creditors would not be bailed in, the central bank decided to bail in subordinated debt holders as a prerequisite for recapitalization after the audit revealed a capital hole larger than initially expected. After failing to reach the necessary quorum in the shareholder meeting to approve this write-down, the central bank requested amendments to the bankruptcy law. Only after the amendment to the bankruptcy law could the authorities impose losses on the subordinated creditors and the existing shareholders. In the interim, as deposit outflows continued at the bank, the central bank had to provide significant liquidity, by placing deposits into the bank to enable the bank to meet its obligations as it awaited recapitalization. After the recapitalization, the bank was able to repay this deposit from the central bank in 2018.

Additionally, clarity and consistency are crucial, as observed in the restructuring and resolution survey. Any unexpected actions by the authorities resulting in unexpected losses to stakeholders can trigger lawsuits and create uncertainty for others and related markets, further destabilizing the financial system. In the case of Portugal's BES (Novo Banco), the authorities relied on the public interest clause to override the contractual claims of some senior bondholders. The authorities retransferred these senior bondholders back to the bad bank so that it could impose larger losses and lower the cost to the government (Gupta and Brandon 2024). However, this led to lawsuits and uncertainty in the senior unsecured debt markets, as investors became concerned that their debt may also become bail-in-able. The resolution and restructuring survey included the most recent example of Credit Suisse in 2023, which further emphasized the importance of clarity and consistency. The authorities in Credit Suisse wrote down additional Tier 1 (AT1) bondholders before the equity holders. Although the terms of the AT1 bonds included such a possibility, the exact process used was surprising to many market participants. Immediately, lawsuits were brought by those AT1 bondholders suffering losses. Also, regardless that the AT1 bonds allowing for such write-down were unique to Credit Suisse, other institutions' AT1 bondholders suddenly became worried about the possible treatment of their holdings, and the market temporarily came to a halt (McNamara et al. 2024).

In the Yes Bank case, although India's authorities had not included in the press release of the bank's recapitalization its plans to write down AT1 bondholders, the central bank did so. It also wrote down only certain AT1 bondholders for undisclosed reasons. The AT1 bondholders who suffered losses brought a lawsuit, and India's High Court ordered a stay on the write-off of AT1 bonds. It reasoned that although the administrator of the bank had the legal authority to order a write-down of AT1 bonds, it did not obtain an approval from the Ministry of Finance, and only some AT1 bonds were written off without adequate reason. The government appealed, and the case is currently at India's Supreme Court (Gupta 2024).

On the other hand, Iceland negotiated agreements with creditors to avoid potential lawsuits from the write-down of EUR 28.1 billion owed to senior creditors, an amount that was twice Iceland's 2007 GDP. Senior creditors received an average 29% recovery for their claims. Under the terms of their agreements with the authorities, creditors would receive compensation from contingent bonds if resolved assets turned out to be worth more than expected. They also received an exemption from capital controls, in return for agreeing not to bring a lawsuit against the government for their losses (George 2024d).

8. Capital Characteristics: What form of capital was injected? What other features did authorities use to allow taxpayers to share in potential losses or gains?

Common shares with voting rights were the most popular form of capital injected by policymakers among the 23 AHCI cases we surveyed. Eighteen cases used common shares. Of the 18, six cases combined common shares with another form of capital. Figure 5 shows that preferred shares were less popular among the AHCI cases we surveyed. The BBCI programs we surveyed earlier were mostly voluntary, requiring the consent of common shareholders, and, for that reason, mostly involved the issuance of new preferred shares rather than common shares.

Figure 5: Capital Characteristics

Case Name	Capital Characteristics			Bail-in
	Common Shares	Preferred Shares	Other Capital	
Austria and Germany, HAA	✓		Participation capital	✓
Belgium and France, Dexia	✓			
Belgium, Fortis	✓			
Cyprus, Laiki	✓			✓
France, Bank of France	✓			
Germany, HSH	✓			
Germany, IKB	✓		Loan with "better fortune" clause waiving repayment obligation	
Greece, ATE	✓			
Greece, Piraeus	✓		CoCos	✓
Iceland	✓		Subordinated debt	✓
India, Yes Bank	✓			✓
Italy, MPS	✓			✓
Japan, NCB	✓	✓		✓
Korea, KFB and SB	✓ Unknown for SB			✓
Latvia, Parex	✓		Subordinated debt	✓
Netherlands, SNS Reaal	✓			✓
Portugal, BES (Novo Banco)	✓			✓
Russia, Otkritie	✓			✓
Spain, CCM		✓		
Switzerland, Volksbank			Cooperative capital	✓
Switzerland, UBS			Mandatory convertible notes	
US, BofA		✓		
US, Citi		✓ Converted to trust preferred		

Source: Authors' compilation.

In some cases, existing law shaped the decision on what form the injected capital would take. For example, as we saw in the BBCI survey, Austrian law stated that in recapitalization by the authorities, the capital injection was to be in participation capital, which was an instrument in Austria to increase an institution's core capital ratio and enhance risk-bearing capacity but did not grant authorities any position or membership rights under corporation law. The preference for participation capital stemmed from the banks' influence on the writing of the law (Decker 2024a). In KFB, the government had initially planned to inject preferred shares. However, Korean commercial law prohibited the issuance of preferred stocks beyond 25% of any firm's total outstanding capital. Thus, the maximum amount the government could have injected would have been less than the intended size of the capital injection. Therefore, the government provided the capital in ordinary shares (Park and Heaphy 2024).

Overall, however, policymakers use security choice and shareholder rights to reinstate market confidence in the institution. In Volksbank, policymakers injected cooperative capital. The central bank initially recommended using preferred shares ranking above the existing cooperative shareholders. The fiscal authority, however, was convinced that it should not create two types of capital with different seniorities, arguing this would create further public mistrust, especially after the existing share value had already been written down in half (Makhija 2024b).

Common Shares

In the case of Latvia's Parex, the policymakers initially injected subordinated loans but changed to common shares for subsequent capital injections after the subordinated loans failed to stem the deposit run in the bank. The IMF, after witnessing the injection of subordinated loans, had criticized that this was a misstep of the policymakers and it would not be enough to stem the run (Decker 2024b).

In the instance of Citi in the US, the policymakers first injected preferred shares into the bank. As discussed in the BBCI survey, during the GFC, US policymakers chose preferred shares rather than common shares owing to concerns about the political ramifications of a large government stake in a private bank's common equity. Additionally, policymakers reasoned that by avoiding market characterization of the bank as having been nationalized, they could maintain market confidence in the bank. Preferred shares also had other advantages: (a) more seniority in the capital stack, resulting in higher protection of taxpayers in the event of liquidation; and (b) the payment of dividends by the bank to the government. However, policymakers later announced that they would exchange the preferred shares into trust preferred securities aimed at boosting the bank's common equity. These securities were convertible into common equity. Since 1996, a portion of these securities satisfied Tier 1 capital requirements in the US under the banking regulatory standards then in place. However, while regulators deemed them equity for regulatory capital, rating agencies rated them as debt instruments. Moreover, the country's deposit insurer had stated that in its experience, the holders of these securities had impeded recapitalizations or sales of troubled banks (Hoffner and Arnold 2024b).

In chronic situations, we still witness more injection of common shares. This is because most of those banks in chronic situations are being resolved and restructured, and often authorities are the sole shareholders of the institution.

Other Features Attached to Noncommon Shares

Some authorities attached specific features to qualify loans or other noncommon shares as Tier 1 capital, using mechanisms through which the government would share in the risk of losses. For example, the UBS case used mandatory convertible notes. These notes would automatically convert to ordinary shares 30 months after issuance, and this mandatory conversion allowed the bank to count these notes as Tier 1 capital from the date of issuance. The bank, however, was required to pay the full coupon payment for the entire term. In a report to the Swiss Parliament, policymakers explained that these notes had the advantage of providing a fair and secure remuneration in the form of the coupon rate and allowed them to avoid becoming co-owners of the bank, so their participation in UBS's capital would not interfere with bank supervision (Makhija 2024c).

Other cases used bespoke loss-sharing features. For example, in the case of Germany's IKB, an official loan included an immediate waiver of the bank's obligation to repay unless the bank's finances improved ("better fortune" clause). The loan also included a waiver of regular interest payments when the bank was incurring accounting losses. IKB's regulator ruled that these waivers enabled the loan to qualify as Tier 1 regulatory capital. IKB also issued another form of hybrid capital called profit-participation certificates, the holders of which were entitled to dividends based on the performance of the issuing company. Payments out of future profits on obligations that used the better fortune clause would precede payments on these certificates (Swaminathan and Vala 2024).

Alternatively, some programs used warrants to make it possible for taxpayers to share in a bank's recovery. In Citi and BofA, the US law authorizing capital injection required authorities to acquire warrants for the right to receive nonvoting common shares or preferred shares of the financial institution for the benefit of taxpayers. Although it was not a condition of the capital injection, Swiss authorities also used warrants as a part of acquiring bad assets of UBS before the capital injection, for a similar purpose as with Citi and BofA.

Capital Features to Promote Exit

Some authorities used features of the capital design to incentivize banks to exit from the aid as soon as viable. In Piraeus, the European Commission stated that the use of contingent convertible (CoCo) bonds ensured the temporary nature of the aid as it was repayable (Schaefer-Brown 2024c). The authorities in Piraeus injected 25% of the capital in common shares and 75% in CoCo bonds. On the other hand, in CCM, the Commission criticized that the low interest rates and long term on preferred shares did not provide a sufficient incentive for the bank to repay the government for the capital injection. The preferred shares paid no interest during the first five years, and afterward, the interest rate was 3% until the 10th year (Swaminathan 2024).

Capital Features to Encourage Private Investment

In other cases, authorities used features of the official capital injection to encourage or require banks to raise private capital. In the BofA case, the authorities stipulated that the bank couldn't repay the government for its AHCI unless the bank had first repaid its capital received from the BBCI program, which the authorities had also provided—but the bank couldn't repay the authorities for either program until it raised USD 25 billion in capital from private investors. In Piraeus, in order for the bank to receive the capital injection, the authorities first required the bank to privately raise enough capital to satisfy the capital needs identified under the baseline scenario of a stress test by the Single Supervisory Mechanism. After a successful private capital raising, the authorities injected the capital needs identified under the stressed scenario.

9. Size and Source of Funding: What was the announced size of the capital injection? Did changing circumstances lead the authorities to increase it? Where did the funding come from? Were the authorities providing cash or bonds in exchange for the capital? Were any private funds involved? What measures did authorities take to protect the provided funds and allow them to recoup their investments?

In discussing broad-based programs, we noted that the overall size of the capital injection is influenced by whether the crisis is in an acute or chronic phase. We pointed out the risk of deploying too little capital in an acute phase, when the authorities are concerned about contagion through bank counterparties, although ultimately the size of the injection is limited by fiscal and political realities. We observed some parallels with this distinction in the ad hoc cases, but with acute and chronic defined at the entity level. In the ad hoc cases, we witnessed the same concern around injection being too small in the BES (Novo Banco), HAA, Parex, Laiki, NCB, Citi, BofA, and KFB and SB cases. In these cases, overoptimistic estimates and the continuous increase in the bank's nonperforming loans (NPLs) were some of the reasons the size of initial injections was insufficient (Heaphy 2024; Park and Heaphy 2024; Schaefer-Brown 2024a).

We also witnessed that political realities influence the calculation of the size of the capital injection in ad hoc capital injections. In ad hoc cases, the authorities often used stress tests, models, or audits to figure out the rough size of the injection. However, the calculations were not without fault. There were disagreements between the European Commission and national authorities in HSH and Laiki as to the size of the losses of the banks. In HSH, the Commission ended up ordering the bank to compensate further the authorities on the support they provided overall, and in Laiki, the bank ended up needing additional capital injection than initially calculated. In Greece's Piraeus, the stress test by the Single Supervisory Mechanism was able to accurately calculate the capital needs of the bank without influence or conflict with national authorities.

Generally, the funding for capital injection came from the fiscal authority, and the authorities could contribute cash or bonds, either tradable or nontradable, to the banks. However, KFB and SB initially used central bank funds and then moved on to deposit insurance funds. CCM used deposit insurance funds. The Otkritie and NCB cases used central bank funds.

Others relied to some extent on funds from a public bank or consortium of private parties. Yes Bank, BES (Novo Banco) , HAA, and IKB had a group of private banks funding part of the capital injection. The central bank coordinated the capital injection from a consortium of banks in Yes Bank. BES (Novo Banco)'s funds came from the resolution fund that had been funded by the state and a group of credit institutions. HAA had existing shareholders contributing additional funding to the capital injection until they were wiped out. IKB recapitalization was funded by a state-owned development bank, with half of the amount guaranteed by the federal government (Swaminathan and Vala 2024).

What banks get in return for the issuance of capital is also important in signaling credible support. Governments sometimes issue bonds to troubled banks, rather than cash, to replenish their capital deficit. Injecting cash is considered preferable (Moretti, Chavarri, and Dobler 2020). However, various political or technical constraints may compel governments to issue bonds to recapitalize a distressed bank. Although we did not study the US Savings and Loan (S&L) crisis in this survey, in 1982, during that crisis, the US Congress was reluctant to take measures that would increase the budget deficit. As a compromise, it authorized the FDIC and the Federal Home Loan Bank Board to administer a temporary bond-based recapitalization program (Allardice et al. 1983). The FDIC invested in net worth certificates (NWCs) with participating banks in exchange for FDIC-insured promissory notes (FDIC 1998, 23). The FDIC considered NWCs to be a form of regulatory capital, which recipients redeemed as they returned to profitability by surrendering an equal portion of FDIC promissory notes. The recapitalization was essentially cashless, unless the recipient bank failed (US GAO 1984), and the program was largely considered successful (FDIC 1997, 230).

Bond-based recapitalizations were a very common practice in financial crises before 2000 (Andrews 2003), although we noted in our own survey on broad-based capital injections that they have been less common since the Global Financial Crisis (Rhee et al. 2022). Of course, this strategy may not work for a country with a poor credit rating. The capital injection did not stop the depositor runs on Laiki Bank in Cyprus. Not only was the injection amount insufficient to absorb the continuing losses from Greek loans but also the ECB noted that the use of an unfunded sovereign bond from that country may not have been viewed as credible and that recapitalization by a government bond could create a feedback loop between the state and the bank. However, Cyprus may not have had a choice as it was not able to access funding markets after rating agencies downgraded its sovereign debt to non-investment grade (Schaefer-Brown 2024a).

When deviating from cash-based recapitalizations, governments should ensure that the terms and conditions of the bonds enable the bank to remain viable. Bonds carrying an embedded sovereign credit risk or paying below-market rates may impede the recipient's profitability and result in a prolonged and costlier resolution process (Andrews 2003). The IMF recommends that governments resorting to bond-based recapitalization make sure that: (a) interest rates are high enough for the recipient bank to remain profitable, (b) maturities are consistent with the expected recovery in the franchise value of the bank, and (c) the bonds are marketable and accepted by the central bank as collateral (Moretti, Chavarri, and Dobler 2020).

10. Timing: Was the program timely?

Delays in intervention can cause further erosion to the capital base of a bank by accelerating runs. In Otkritie, the capital injection took place over three months after the announcement, in part due to the authorities having to wait for the legal amendments allowing for bail-in. In the meantime, the central bank placed the Otkritie in temporary administration and provided extensive liquidity support to meet continued deposit withdrawals before the recapitalization, after which the deposit outflow reversed (Hoffner 2024b). In UBS, although supervisors were monitoring the bank and knew of significant troubles in the bank, they did not notify the government as they were concerned about information on the troubles of the bank leaking. A lack of trust among authorities led them to not deal with the financial crisis for five months. After the bankruptcy of Lehman Brothers in September 2008, the situation at UBS deteriorated rapidly, and UBS officially requested aid from the Swiss government on October 14, 2008 (Makhija 2024c).

In other cases, the authorities were able to respond in a timely manner. In the BofA case, authorities announced recapitalization in time for the bank's disclosure of losses it had incurred from a merger with a troubled bank (Hoffner and Arnold 2024a).

Sometimes, capital injections took place after private capital-raising efforts failed (MPS, SNS Reaal, Laiki, UBS, Yes Bank). In other cases, liquidity support, guarantees, and other measures proved insufficient, and the authorities chose to inject capital (ATE, Volksbank, IKB, HSH, KFB and SB). In the cases of Citi and BofA, the broad-based capital injection they had received in late 2008 was not enough, and they received another ad hoc capital injection from the authorities (Hoffner and Arnold 2024a; Hoffner and Arnold 2024b).

Some cases were chronic phase capital injections where there was enough time to perform a comprehensive assessment and stress tests on capital shortfalls. Also, the bank had enough time to restructure and try private capital raising first. Piraeus was in this situation and raised enough from private investors to cover the capital needs under the baseline scenario of the stress test, and the authorities made up for the shortfall from the adverse scenario (Schaefer-Brown 2024c). Authorities in HAA, Parex, IKB, Iceland, BES (Novo Banco), and CCM continued to or started to inject capital as the banks entered resolution.

Compared to the BBCI cases, regulations around shareholder protection were not as large a hurdle in ad hoc cases. This may be because the assistance was individually targeted and most firms were in absolute need of assistance at the time of capital injection or they had already gone through resolution when they reached the point of capital injection. In all of the AHCI cases, participation in the interventions was mandatory—the banks did not have another choice, and the government had decided it had the authority to require the company to accept the capital—whereas most of the BBCI programs were voluntary.

11. Restructuring: What restructuring did the bank have to commit to as a condition of receiving capital injection?

Similar to the BBCI programs, restructuring paved the way toward long-term viability and was politically beneficial for ad hoc capital injection programs. More than half of the ad hoc capital injection cases required restructuring plans as a condition for receiving aid, mostly because of the imposition of conditions by the European Commission. Conditions included divestment of entire businesses (SNS Reaal, NCB), disposal of selected portfolios including nonperforming loans and reduction of the size of a balance sheet (MPS, Volksbank, UBS, HSH, IKB, NCB, BES [Novo Banco]), and other cost-cutting measures (reduction in operating expenses by cutting head count, branches, and other administrative expenses) (MPS, Piraeus).

12. Treatment of Board and Management: Was the board or management replaced as a condition of the capital injection?

Replacing the board and management serves a similar purpose as restructuring the bank for long-term viability and political benefits. Some authorities forced the resignation of the board or management and appointed replacements (SNS Reaal, Yes Bank, Laiki, IKB, Iceland, BES [Novo Banco], NCB). One appointed a trustee to monitor the activities of the board (ATE), and another appointed a new board member to the existing board without replacing any of the old members (Piraeus).

Volksbank had harsher requirements where a condition of the government's capital injection was that the previous management of the bank should be examined and, if necessary, civil and criminal charges should be pursued against them (Makhija 2024b). In other cases, the board resigned voluntarily (UBS, HSH, Fortis, Dexia, HAA, Parex) and waived compensation (UBS).

In the US cases, the right to appoint directors to the board was contingent upon the banks' payment of dividends on the preferred shares issued to the government. If the bank missed dividend payments for a total of six quarterly dividend periods, the government could appoint two directors to the board.

However, having the authorities appoint directors to the board may lead to some governance problems. In Korea's SB, to improve the governance structure and mitigate moral hazard, the new board was structured so that outside directors outnumbered insiders, and an audit committee with a standing auditor was incorporated. However, filling the positions in the new board during a crisis proved difficult. SB lacked consistent management, having three interim CEOs and one regular CEO from 1997 to 2000. Some attest that while this structure may have alleviated the political concern around using public funds, it still was not enough to serve the functional needs of bank governance. Although the outside directors were well-informed and committed, the policymakers who were the new shareholders did not take the board seriously, according to Kang (2003). Moreover, outside directors were not included in the discussion of the status of the bank in general or the timing of the sale, as shown in unsuccessful attempts to sell SB (Kang 2003).

13. Other Conditions: Did recapitalization come with nonprice conditions to incentivize the banks to exit the government support quickly?

Nonprice conditions incentivize banks to exit quickly and can also be politically beneficial. Restrictions on compensation (SNS Reaal, MPS, UBS, HSH, HAA, Iceland, BES [Novo Banco]) and dividend payments (MPS, Laiki, Iceland, BES [Novo Banco]) were widely used nonprice conditions for the AHCI cases, as in the BBCI cases. Other conditions included restrictions on a bank's business activities (SNS Reaal, ATE, Laiki), making coupon payments (SNS Reaal, MPS, Laiki), and advertising government support (SNS Reaal, BES [Novo Banco]) to ensure that a bank wasn't taking advantage of government support to compete with other banks or take excessive risks.

Compared to the EU countries, the US did not impose an up-front restriction but a requirement to obtain consent or approval for senior compensation plans and paying dividends.

Owing to a unique situation of recapitalization by a consortium of banks, in the Yes Bank case, the major nonprice condition was that each bank participating in the consortium had to hold at least 75% of its holdings for three years after the injection. No taxpayer funds were invested in this case, and perhaps the need for stability trumped a need for quick exit by the bank receiving the injection (Gupta 2024).

14. Regulatory Relief: Did the authorities amend or grant exemptions from any existing regulations to assist smooth operation of the capital injection?

For the BBCI programs, much of the focus of this section was on incentivizing participation in the program. Since the AHCI cases were mandatory, such incentives were not needed. Instead, our AHCI cases noted examples of authorities allowing for exemptions from the existing regulatory requirements to smooth the operations of the capital injection.

In Yes Bank, the authorities granted exemptions for the banks participating in the capital injection from the capital gains tax on profits from the sale of shares after the lock-in period (Gupta 2024). Also, other common exemptions we saw were the continued operation of the bank during the capital injection, even if the regulatory capital ratio was breached (Yes Bank, Otkritie, Iceland's three banks).

15. Exit Strategy: Did the authorities have a plan to unload their investments once conditions normalized? Was the plan in place at the time of capital injection?

Authorities generally prefer to exit the investments in AHCI programs as soon as possible. To achieve a successful exit, having a plan already in place at the start is helpful. The European Commission often supports and emphasizes quick exits for the authorities. Having a goal date to privatize alone is often not enough. Many ad hoc cases showed that too often these goal dates were passed (MPS, BES [Novo Banco], KFB and SB) because of a lack of interest from private institutions or failed negotiations with private counterparties in the sale of shares authorities owned. In MPS, the authorities agreed to sell off their stake by the end of 2021, but as of November 2023, they still owned 39.2% of the bank (Hoffner 2024a). BES (Novo Banco) received EUR 7.3 billion of capital injections over eight years, and the authorities were set to exit by the end of 2016. However, they still had a 25% stake in the bank with a book value of EUR 1.1 billion as of 2023 (Gupta and Brandon 2024). A more concrete premeditated plan is helpful, but the majority of our cases lack the plan.

Interestingly, on the other hand, plans also need flexibility, and having a goal date could backfire under certain circumstances. In KFB's case, Korean authorities had set up a plan for privatization in advance with the preference to sell the bank as a whole, but if there were to be a split sale, it had to be one buyer purchasing 50%+1 share so that the buyer would secure management rights while the authorities could maximize the opportunity of recovering taxpayers' money. Amongst the two bidders they had for the bank, the authorities preferred one willing to purchase 50%+1 share instead of 80%. Moreover, the existence of a deadline for privatization created speculation that the government accepted the deal to meet the deadline set by the IMF. The deal provoked a public outcry over "fire sale" privatization of the bank (Park and Heaphy 2024). While the structure of the exit plan was motivated by good goals, the political costs exceeded the benefits of such goals.

At the time of our survey's publication, governments still retained a stake in seven of the financial institutions profiled in our survey.

Conclusion

This paper surveys 15 Key Design Decisions for 23 ad hoc capital injection cases. Our close examination of these cases reinforces the themes of our earlier broad-based survey while also providing insights into the write-downs and bail-ins more common in the ad hoc examples.

To be sure, capital injections can be costly, financially and politically. Nonetheless, authorities often find that they are necessary to protect depositors and prevent financial contagion. Countries with relatively robust supervisory and legal authorities are more likely to take successful preemptive actions to address poorly capitalized banks. Authorities should be aware of regulatory constraints on program implementation. These constraints are often far from the top of policymakers' concerns but ignoring them can lead an otherwise well-designed program into unnecessary delays. Key Design Decision No. 3, Legal Authority, and Key Design Decision No. 10, Timing, discuss some examples. As emphasized in our consideration of broad-based cases, policymakers are encouraged to address as many of these problems as possible through legislation passed in quieter times before a crisis occurs, allowing for emergency authority to be utilized efficiently.

BBCI Themes Reinforced by Ad Hoc Capital Injection Cases

The size of a capital injection must be "large enough" to solve the problem without the need for further injections. In the ad hoc cases, we witnessed the same concern around injection being too small in the BES (Novo Banco), HAA, Parex, Laiki, NCB, Citi, BofA, and KFB and SB cases. In these cases, overoptimistic estimates and the continuous increase in the bank's NPLs were some of the reasons the size of initial injections was insufficient (Heaphy 2024; Park and Heaphy 2024; Schaefer-Brown 2024a).

Capital injections should be combined with other interventions. Similar to broad-based programs, combining other programs with an ad hoc capital injection can be a useful force multiplier. Although capital injection is a powerful intervention tool, it is politically and fiscally expensive. As discussed above in Key Design Decision No. 2, Part of a Package, guarantees can especially be a powerful force multiplier. Fourteen of the 23 ad hoc cases included guarantee programs. The type of guarantee varies depending on the failing bank's situation. Guaranteeing a bank's liabilities may help stem runs by anxious creditors and depositors, but it may not convince anyone that the bank itself is solvent. For that reason, in many cases, governments have also guaranteed a bank's assets, capping the potential for further credit or other losses. In the HSH case, German authorities, after guaranteeing banking debt, combined a capital injection and an asset guarantee to reduce the bank's risk-weighted assets. The purpose was to ensure the bank met regulatory capital requirements while minimizing the budgetary impact of the injection. In the capital injection into Citi, US authorities constructed a package that consisted of both a second round of public sector capital and an asset guarantee; they had already guaranteed all bank liabilities. This was in response to the market's growing concern about the quality of Citi's assets.

Beware of legal or regulatory constraints on program implementation. In 15 of the AHCI cases, the capital injections relied on new legal regimes. This meant that some authorities needed to be creative in stretching existing regimes to find a basis for a timely capital injection. Swiss authorities in the UBS case relied on an interpretation of existing government authority to issue ordinances to protect Switzerland's foreign relations and security, and the Swiss Parliament approved the capital injection retroactively. Subsequent legal analysis has shown that this interpretation was a stretch (Makhija 2024c). Without appropriate legal authorities bestowed upon the policymakers, it took months for the authorities in the Otkritie case to write down subordinated creditors and shareholders, delaying the capital injection. In the interim, Russia's central bank had to provide liquidity as the bank's deposit outflows continued (Hoffner 2024b).

Exit plans are important, and capital design can help with the exit. As we noted in the BBCI survey, the absence of a feasible exit plan can lead to higher costs, both politically and economically. Although almost all the authorities in the 23 AHCI cases we surveyed declared their intent to reprivatize as soon as possible, very few had a feasible exit plan. In 11 of the 23 cases surveyed, the authorities ultimately put the financial institution receiving capital injection into resolution. In 12 of the 23 cases, the authorities eventually found private sector buyers, including the banks that had gone through resolution. In the Otkritie case, a state bank bought the authorities' stake in the bank. At the time of our survey's publication, governments still retained a stake in seven of the financial institutions profiled in our survey.

It may be easier to evaluate the up-front expense of capital injection to the taxpayers—cash invested in bank capital, for example—than to evaluate the ultimate *net* cost, including income from interest or dividends and from the ultimate sale of the stakes or other assets back to private investors. Asset values tend to gradually recover in the years after a crisis, allowing the authorities to recover at least a portion of their investment in a recapitalization. The US government made net profits of USD 4 billion from the USD 40 billion ad hoc capital injections it made to Citi and BofA in the AHCI cases we studied, when losses turned out to be far lower than experts forecast at the height of the crisis. The Swiss government recapitalized UBS for 6 billion Swiss francs (CHF) in December 2008 and by August 2009 sold its stake for CHF 7.2 billion. Iceland's rescue of three banks cost multiples of GDP up-front, but by some measures, the government got all its money back, after imposing severe losses on foreign creditors. (These net cost estimates typically do not include opportunity costs.)

Key Design Decision No. 8, Capital Characteristics, and Key Design Decision No. 15, Exit Strategy, discuss examples helpful in accomplishing the exit.

In the acute phase, higher-tier capital is better. In the BBCI survey, we emphasized that distinguishing between the goals of BBCI programs implemented during acute ("panic") and chronic ("debt overhang") phases of a crisis is crucial. In an acute phase, we found that BBCI programs should be designed to influence the behavior of bank counterparties, while in chronic phases, the focus should be on bank behavior itself. We observed some parallels with this distinction in the ad hoc cases, but with acute and chronic defined at the entity level. When authorities are motivated by liquidity concerns—in the midst of a run or in fear of one—we consider that an acute-phase intervention, even if the overall financial system is not in danger. On the other hand, when authorities are worried more about long-run viability than short-term liquidity, we consider that to be a chronic-phase intervention. In the acute phase for ad hoc cases, the design of capital injection should aim to restore and maintain bank counterparties' confidence. Authorities in the Parex and Citi cases changed the capital characteristics midway into those injections to have a higher tier and a simpler form after realizing that the markets were unsatisfied and that the runs continued with the initial form of injected capital. More examples are discussed in Key Design Decision No. 8, Capital Characteristics.

New Themes from the Study of Ad Hoc Cases

The major challenge in making capital injections is balancing the tension between financial stability measures and the political backlash and moral hazard they may engender. Moral hazard concerns often drive punitive ex ante conditions for capital injections; yet, financial stability concerns, focused on the spillover effects of a large bank failure on individuals and markets, should outweigh moral hazard concerns in the acute phase of a financial crisis. Authorities should instead address the political backlash and moral hazard concerns with tough measures such as bail-in—imposing losses on shareholders and creditors—and discipline for management, including through legal channels if laws are broken. But authorities often balk at such measures in the depth of a crisis. Especially in operating bail-ins, even with appropriate legal authority, clarity and consistency are important factors in implementing the bail-in or write-down of existing stakeholders. Unexpected losses imposed on stakeholders can lead to panic in the wider market and multiple lawsuits against the state by stakeholders seeking compensation. For example, the authorities in the Credit Suisse case in 2023 wrote down AT1 bondholders before equity holders. Although that outcome was clearly described in the AT1 bond documents, some bondholders immediately sued, believing they ranked senior to equity holders based on the traditional treatment of senior debt in the capital structure. Meanwhile, the broader AT1 market temporarily froze as investors in AT1 bonds issued by other institutions, which did not have such terms, suddenly became worried they might receive similar treatment (McNamara et al. 2024).

In contrast, in a chronic situation, there may be less concern that such a surprise could spook investors, and authorities may be more able to impose punitive outcomes on stakeholders. Iceland was able to design an agreement with the senior creditors of its failed large banks, while aggressively writing down the value of their holdings, to prevent them from bringing lawsuits against the state. Nevertheless, political obstacles remain in chronic situations as well, as seen in Italy's MPS case where the bail-in-able instruments ended up in the hands of the retail investors.

Governments may face constraints that render conventional cash-based recapitalizations difficult. Using government bonds in such circumstances can work, provided that such bonds have stable sovereign credit status. Governments sometimes issue bonds to troubled banks to fill the holes on the asset side of bank balance sheets. These bond-based recapitalizations were a very common practice in financial crises before 2000 (Andrews 2003), although we noted in our own survey on BBCI programs that they have been less common since the GFC (Rhee et al. 2022). Governments may choose bond-based recapitalizations because their domestic bond markets are relatively undeveloped, but even countries with developed markets have chosen this path because of political or technical limitations of financing cash-based capital injections. Of course, injecting cash is typically preferred to issuing government debt in return for new bank capital in a jurisdiction where the sovereign debt rating has become unstable. For example, the capital injection did not stop the depositor runs on Laiki Bank in Cyprus. The injection amount was clearly insufficient to absorb the continuing losses from Greek loans, but the ECB also noted that market participants may not have viewed the use of an unfunded sovereign bond as credible, considering that rating agencies had repeatedly downgraded Cyprus's sovereign credit ratings. However, Cyprus also did not have a choice as it was strapped with cash owing to its inability to access the funding market. Key Design Decision No. 9, Size and Source of Funding, discusses the issue in further detail.

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